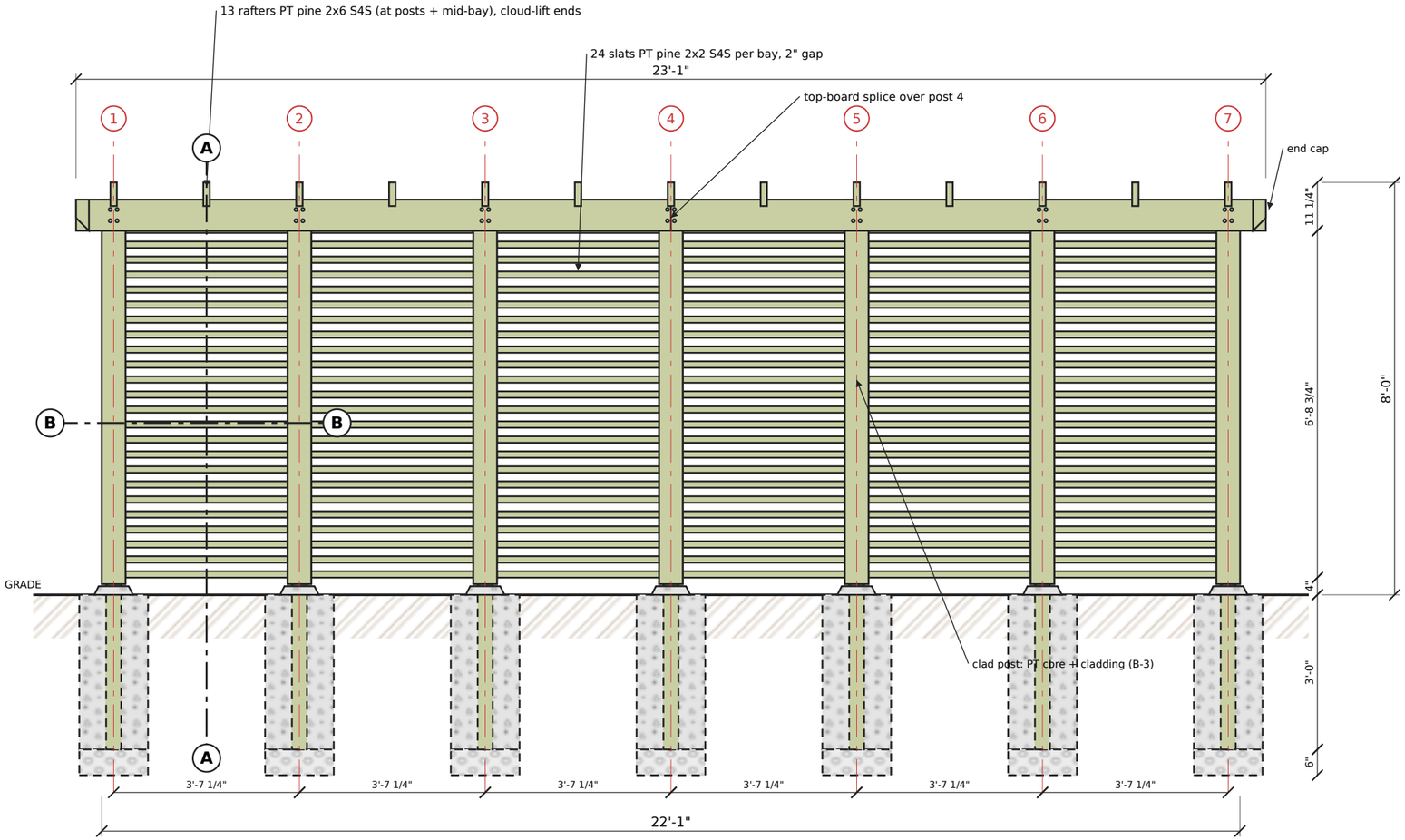
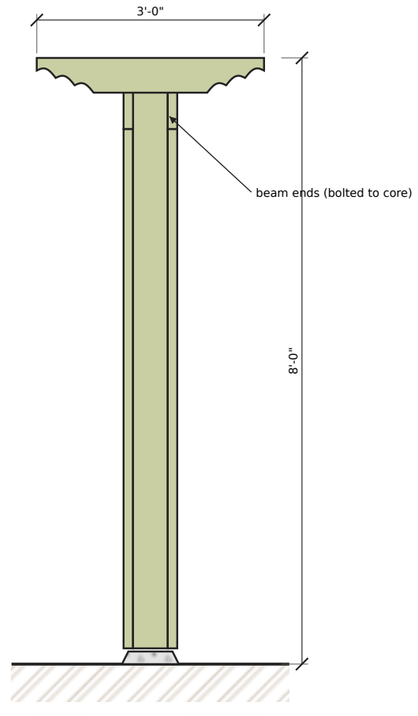


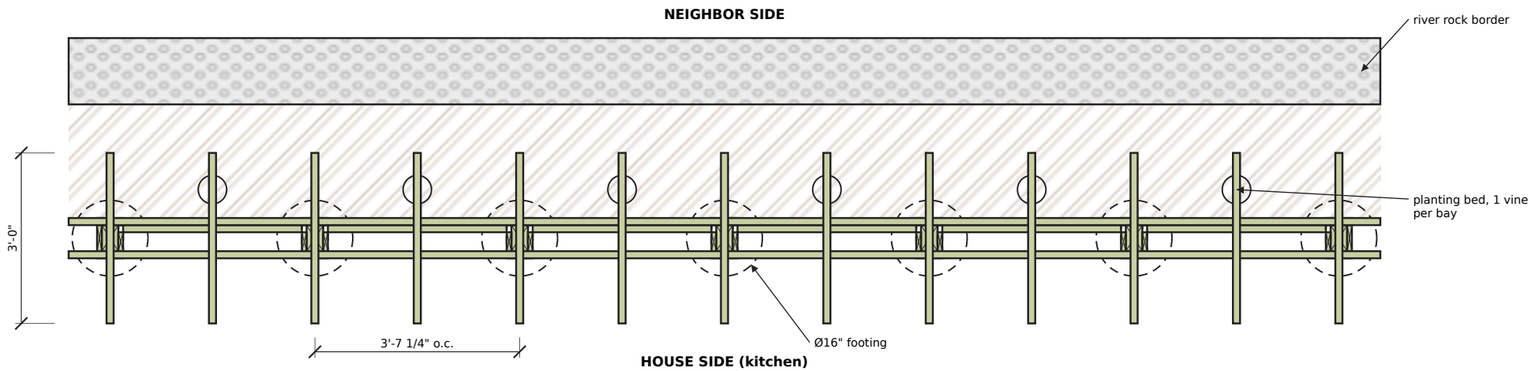
ELEVATION — NEIGHBOR SIDE

SCALE 1:32 (3/8" = 1'-0")



SIDE ELEVATION (end view)

SCALE 1:24 (1/2" = 1'-0")



PLAN — TOP VIEW

SCALE 1:32 (3/8" = 1'-0")

NOTES

1. Overall length 22'-1"; 7 posts at 3'-7 1/4" o.c.; clear bay 3'-1 3/4".
2. Bay/post = 6.86 ≈ φ^4 (Fibonacci 89/13); span 3'-1 3/4" > 36" limit (< 48" max).
3. Post cores embedded 3'-0" in concrete over 6" gravel; holes by machine auger.
4. Call NC811 before digging. Verify property line, setbacks and HOA.
5. Fasteners: coated trim-head deck screws (ACQ-rated), countersunk.
6. 24 slats per bay, 2" gap set by the notched side boards.
7. All posts 5 1/2" wide, faces flush. Parts marked YES are pre-cut in the shop.
8. All lumber: Southern Yellow Pine #2, pressure-treated, S4S; cores ground contact (UC4B).

Kitchen Privacy Screen — 7 posts, pressure-treated pine, 8 ft, pergola top (Cornelius, NC) — Bill

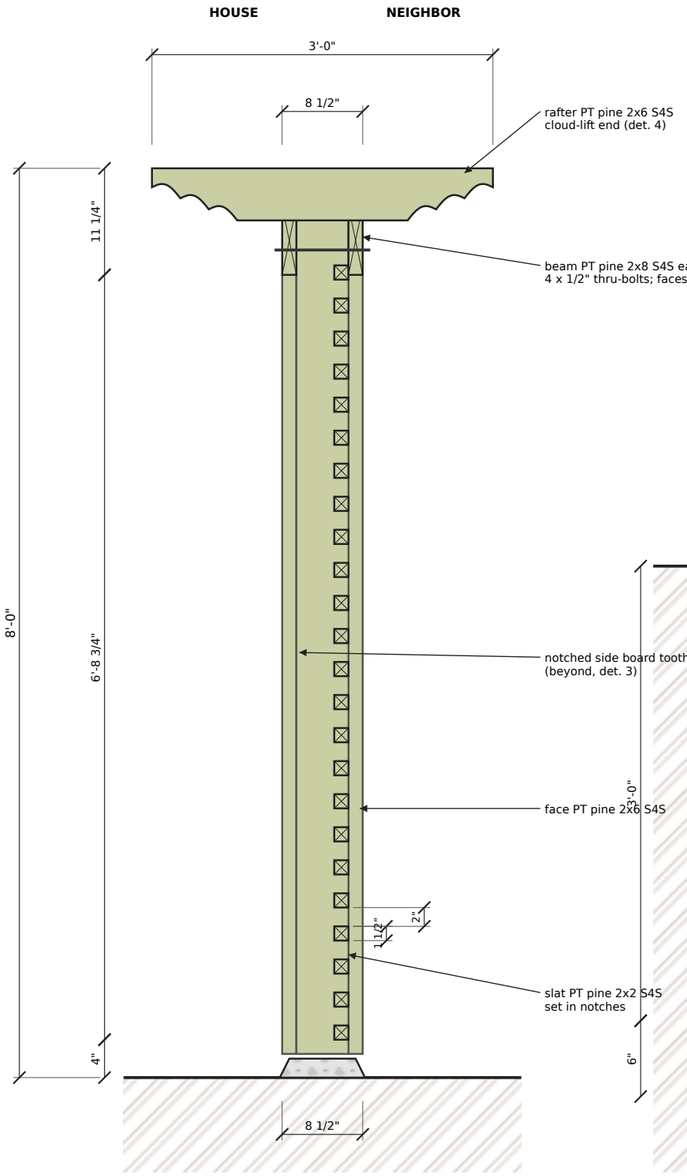
Elevation and plan

Date: 2026-11-10 Rev.: 9 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

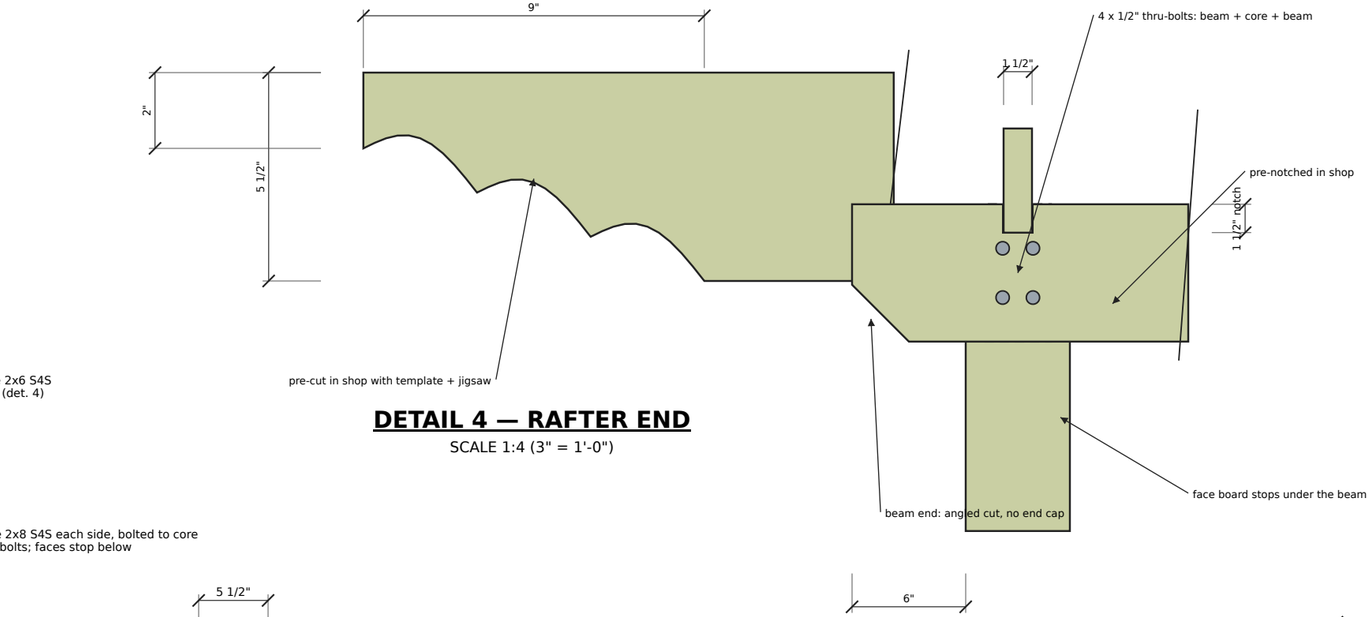
Verify all dimensions in the field before cutting. Not a structural design.

B-1



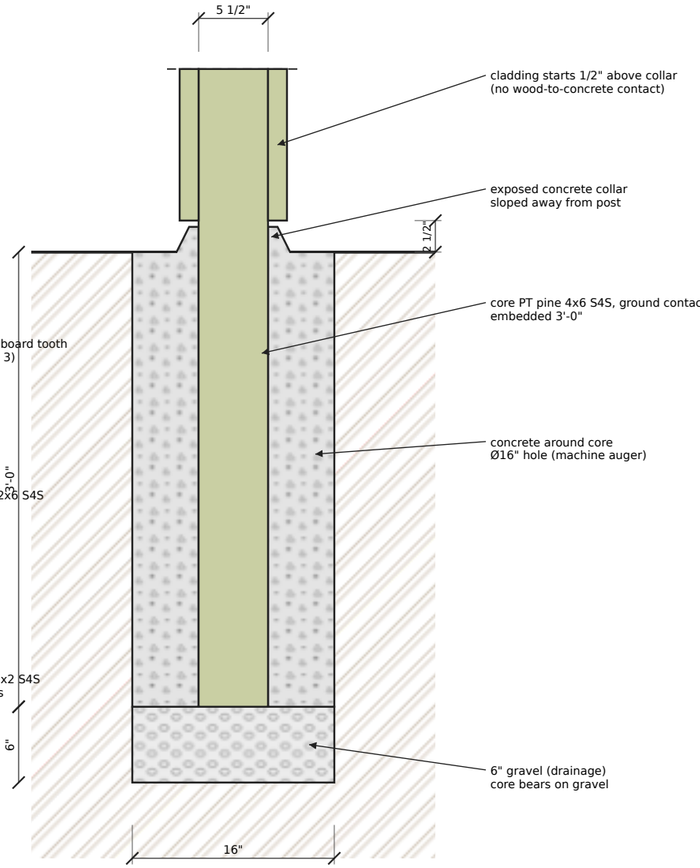
SECTION A-A (mid-bay, looking at post 1)

SCALE 1:16 (3/4" = 1'-0")



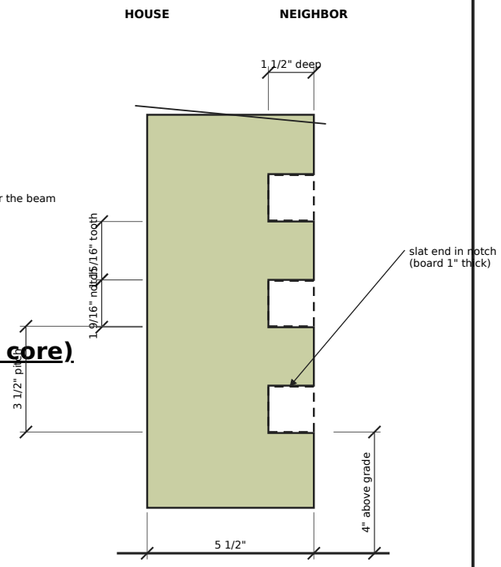
DETAIL 4 — RAFTER END

SCALE 1:4 (3" = 1'-0")



DETAIL 1 — POST FOOTING

SCALE 1:12 (1" = 1'-0")



DETAIL 3 — NOTCHED SIDE BOARD (from bay)

SCALE 1:5

DETAIL 5 — BEAM AT END POST (bolted to the core)

SCALE 1:8 (1 1/2" = 1'-0")

PARTS LIST

Mark	Part	Material	Qty	Length	Shop pre-cut / notes
N	Post core	PT pine 4x6 S4S, ground contact (UC4B)	7	10'-8"	embed 3'-0"; trim tops on site
S	Notched side board	PT pine 5/4x6 S4S notched	12	7'-5 1/2"	YES: 24 notches
R	Slat	PT pine 2x2 S4S	144	3'-3 1/2"	YES: cut to length (stop block)
F	Post face	PT pine 2x6 S4S	14	6'-10 1/4"	YES: cut to length (stops under beam)
L	Plain side (end posts)	PT pine 5/4x6 S4S plain	2	7'-5 1/2"	YES: cut to length
T	Top board (2 per side)	PT pine 2x8 S4S	4	11'-6 1/2"	YES: notches for rafters
K	Pergola rafter	PT pine 2x6 S4S	13	3'-0"	YES: cloud-lift ends

Kitchen Privacy Screen — 7 posts, pressure-treated pine, 8 ft, pergola top (Cornelius, NC) — Bill

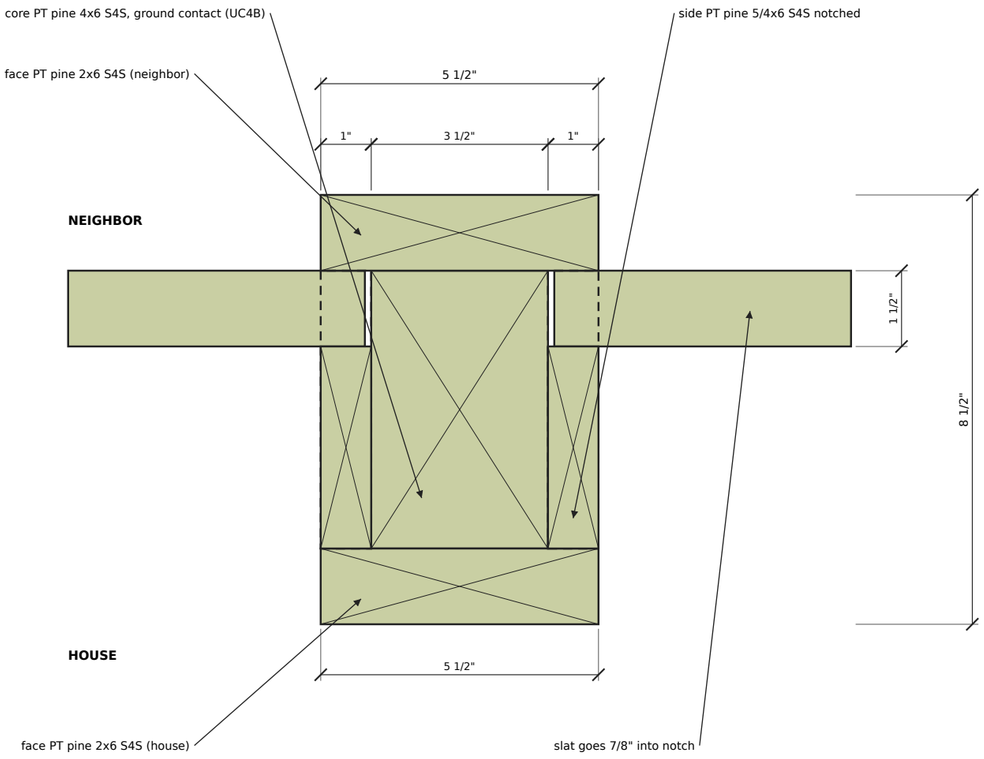
Sections and details

Date: 2026-11-10 Rev.: 9 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

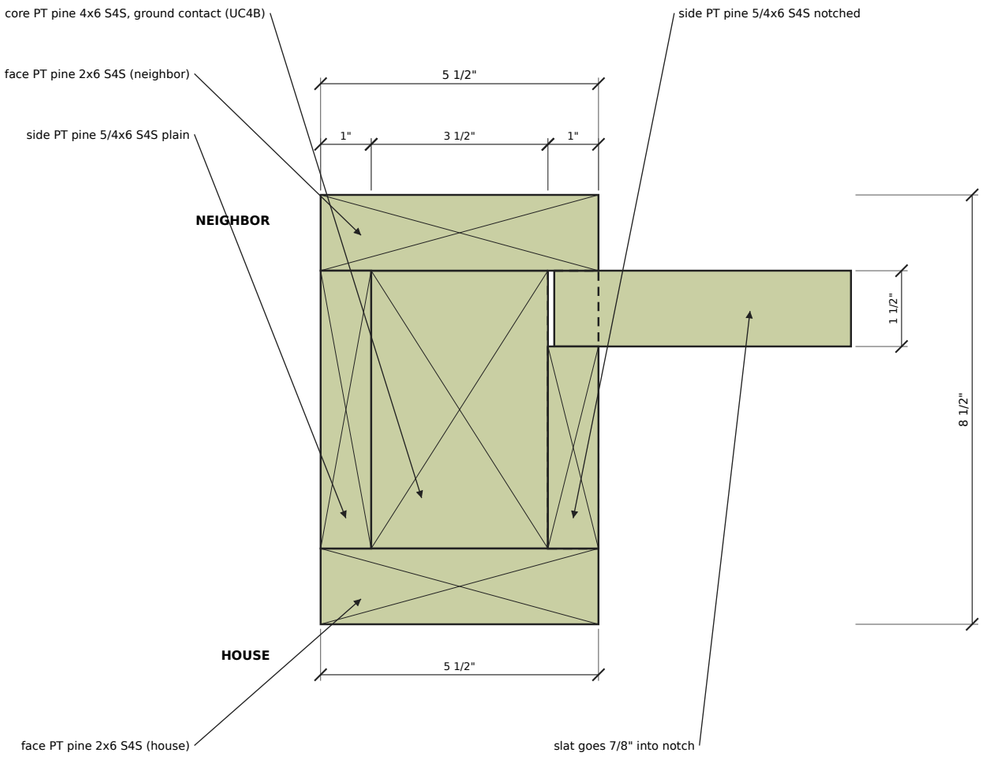
Verify all dimensions in the field before cutting. Not a structural design.

B-2



DETAIL 2a — INTERMEDIATE POST, SECTION B-B

SCALE 1:3



DETAIL 2b — END POST (1 and 7)

SCALE 1:3

#	INSTALLATION SEQUENCE
1	HOLES: lay out 7 post centers at 3'-7 1/4" o.c.; dig Ø16" x 3'-6" with rented machine auger; 6" gravel.
2	POSTS: set PT pine 4x6 S4S, ground contact (UC4B) cores 3'-0" deep, plumb and brace, concrete with sloped collar. Cure 48 h.
3	LEVEL TOPS: cut all cores to 7'-8" above grade (laser level).
4	NOTCHED SIDE BOARDS (pre-cut): fasten to core sides from the neighbor side, notches facing neighbor.
5	SLATS (pre-cut): drop into notches from the neighbor side, bottom to top; 1 screw per end.
6	FRONT CLOSURE: PT pine 2x6 S4S face boards on both faces (closes the notches and locks the slats).
7	SIDE CLOSURE: PT pine 5/4x6 S4S plain on the outer side of end posts 1 and 7.
8	BEAMS (pre-notched): PT pine 2x8 S4S each side, bolted straight to the core above the face boards, 4 x 1/2" thru-bolts per post; splice over post 4.
9	PERGOLA RAFTERS (pre-cut): 13 into top-board notches (posts + mid-bay); hidden trim-head screws.

Kitchen Privacy Screen — 7 posts, pressure-treated pine, 8 ft, pergola top (Cornelius, NC) — Bill

Clad post (PT core + cladding) and installation

Date: 2026-11-10 Rev.: 9 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

Verify all dimensions in the field before cutting. Not a structural design.

B-3